

Benchmarking recruitment rates for phase III trials

Phase III trials that provide the pivotal evidence of efficacy and safety to support the regulatory approval of new drugs represent a major component of the overall cost of drug research and development. Consequently, it is crucial to optimize their operational efficiency. For example, if a phase III trial is estimated to cost US\$150 million over three years, then \$1 million a week is effectively at risk if a trial is delayed owing to operational issues.

Insufficient recruitment of participants has been identified as the most common reason for premature discontinuation of randomized clinical trials, for reasons including over-optimism in recruitment and/or too few recruiting sites planned (*Trials* 22, 844; 2021). With the aim of providing benchmarks for sponsors to set realistic expectations for their business and their trial sites, this article analyses recruitment rates based on data from ClinicalTrials.gov for 2,140 industry-sponsored phase III trials completed since 2011 (see Supplementary information for details).

Recruitment rates

We focused on trials with an enrolment size of 100–2,000 patients, reflecting the most common sizes of interventional phase III trials, with the aim of providing optimal

industry benchmarks. Trials were segmented into four size groups: 100–500 patients, 501–1,000 patients, 1,001–1,500 patients and 1,501–2,000 patients. Trials were also segmented by the type of masking (single, double, triple and quadruple), whether they were for oncology or non-oncology indications, and whether they were sponsored by a top-20 pharma company or not.

Figure 1 shows the recruitment rates for these segments. The average recruitment rate for the trials analysed was 0.84 patients per site per month (ppsm). Trials with enrolment targets between 500–1,500 patients had rates of ~0.57 ppsm, while trials below 500 had rates of 1.05 ppsm and those above 1,500 had rates of 0.28 ppsm. For masking, apart from single-masked trials, the other three types (double, triple and quadruple masked) had a similar recruitment rate, in the range 0.80–0.85 ppsm. Non-oncology trials had slightly faster recruitment rates (0.85 ppsm) than oncology trials (0.77 ppsm). Trials conducted by top 20 pharma companies generally had slower recruitment rates, at 0.62 ppsm compared with 1.00 ppsm for other companies.

Site selection

Trial operations depend on the performance of clinical trial sites. Per the above segments,

trials were annotated by the number of sites listed in the 'Locations' field in ClinicalTrials.gov. Trials with greater than 5 sites listed were selected in order to harmonize the data, as it is improbable that trials with greater than 100 participants are running at less than 5 sites.

As shown in Fig. 2, the average number of sites is generally proportional to enrolment targets, with trials having enrolment targets of 100–500, 501–1,000, 1,001–1,500 and 1,501–2,000 patients involving 55, 105, 144 and 175 sites on average, respectively. Single-masked trials involve an average of 47 sites, while double-, triple- or quadruple-masked trials are conducted across 78 sites on average. Oncology trials were conducted in 71% more sites, at 127, compared to non-oncology trials at 74 sites. Furthermore, top-20 pharma companies conducted trials at 42% more sites, at 98 sites, than all the other companies, at 69 sites.

Actual and predicted enrolment

Trial entries can be tracked to their first version of record in the 'Record history' section of ClinicalTrials.gov. As an example of subgroup analysis, to understand whether or not trials met their predicted enrolment and duration targets, the above trials were further filtered for age (adults, older adults), quadruple-masking, top-20 pharma companies and estimated/predicted enrolment of 500–1,500 patients. This subgroup of 55 trials encompasses several areas of interest: large trials, the most stringent type of trials and the ones with the most financial support. Figure 3 shows the scatter plot of the percentage difference between actual and predicted enrolment and duration of the 55 trials. Positive percentages indicate that the actual enrolment or duration was greater than predicted, while negative percentages indicate the opposite.

Enrolment is largely within a 20% band of predicted targets, although some trials were well below and well above the 20% threshold, suggesting that enrolment targets were changed during the trial for higher or lower numbers of patients than originally intended.

However, the differences between the predicted duration and actual duration (the study start date to the primary completion date) varies

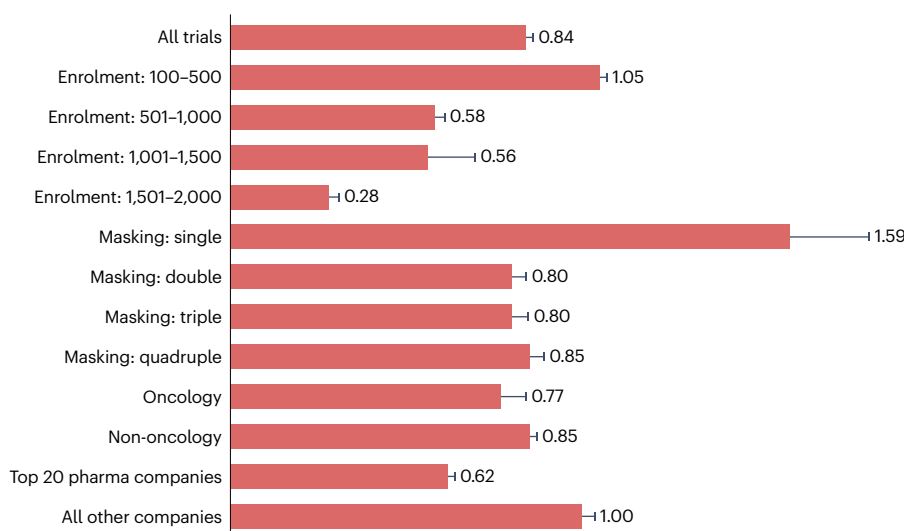


Fig. 1 | Phase III trial recruitment rates (patients per site per month). Trials were segmented across enrolment quartiles, masking type, oncology compared with non-oncology and top-20 pharma companies compared with others. Error bars are standard error of the mean. See Supplementary information for details.

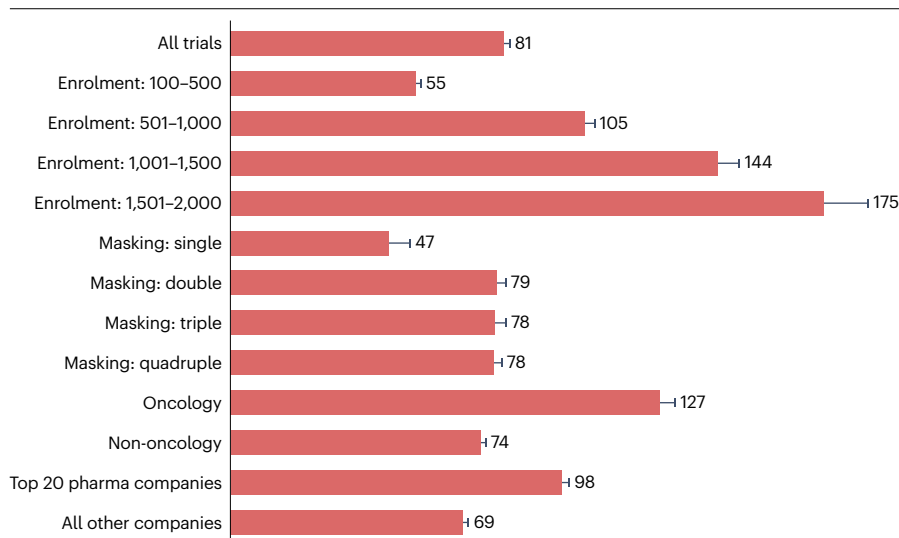


Fig. 2 | Average numbers of trial sites. Sites were segmented across enrolment quartiles, masking type, oncology compared with non-oncology and top-20 pharma companies compared with others. Error bars are standard error of the mean. See Supplementary information for details.

considerably among the 55 trials analysed. More than half the trials had positive percentages, with a very broad range in variance. Some trials took over twice the time to complete. Trials with negative percentages, indicating trials took less time than predicted either due to lower enrolment targets or faster recruitment efforts, also had substantial variance.

So, the analysis indicates that enrolment targets for the 55 trials were generally achieved

by right-sizing the duration during the trial. This is notable, as completion dates are one of the most scrutinized trial metrics because of the heavy dependence of business operations and expectations on when trials actually complete.

Summary

This article presents a methodology to analyse clinical trial operational benchmarks

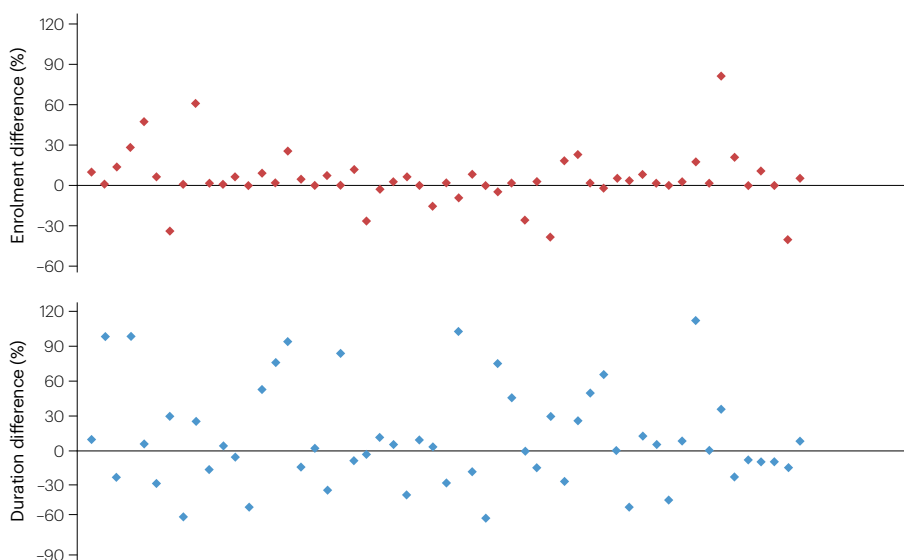


Fig. 3 | Variations in enrolment and duration. The graphs show percentage differences between predicted and actual enrolment (top) and predicted and actual duration (bottom) for a subset of 55 trials (those with actual enrolment between 500–1,500 adult patients and quadruple masking, conducted by top-20 pharma companies).

using the most trusted resource across the industry, ClinicalTrials.gov. Trial sponsors can replicate the analysis for their indication, trial design and/or type of company, in order to benchmark their current trials to those shown to have operational success. This type of analysis is critical at the feasibility stage, when initial discussions with sponsors and contract research organizations are occurring, to set primary completion dates and help ensure trials meet these deadlines. A subgroup analysis of actual compared with predicted targets is also recommended to understand the variance in metrics that can occur during the course of a trial and set up appropriate mitigation efforts.

The benchmarks presented here could serve as a starting point for broader and deeper analysis at the indication, trial design and type of company levels. A subgroup analysis that segments trials with enrolment between 100–1,000 patients, by type of company, and across oncology and non-oncology indications is also provided in Supplementary information. This segmentation provides a starting benchmark to right-size trials to the type of company and therapeutic area. If any of the parameters differ from these benchmarks, the onus should be on the trial operators to justify how and why their trial would exceed historical averages, especially when similar trials have been completed by top-20 pharma companies, given their reach and resources.

Regular surveillance analysis at the trial level can also be performed once trials are in progress to track site performance, staff operational burden and operational expectations from trial start to completion. De-risking \$1 million a week through effective operational management could have a major impact on market entry timelines, business strategy and investor relations.

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Competing interests

The author declares no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/d41573-024-00183-4>.